

NGUYEN THANH PHAT

Tech Lead / Full-stack Developer — Node.js · React · PHP

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SUMMARY

Tech Lead / Full-stack Developer with **~10 years of experience** (since 2016; Tech Lead since 2022) building enterprise back-office, IoT and SaaS systems, including a platform that manages and controls **250,000+ network devices** and one serving **5,000+ students and lecturers**. Strong in TypeScript (NestJS, Next.js/React) and PHP (Phalcon) with PostgreSQL, MongoDB, Redis/BullMQ, RabbitMQ and Kafka; hands-on with SSO/IAM (Keycloak, OAuth2/OIDC), SpiceDB (ReBAC) authorization, multi-tenant architecture and Dockerized environments for the full stack.

SKILLS

Languages: TypeScript, JavaScript, PHP

Frontend: React, Next.js, Tailwind CSS

Backend: Node.js, NestJS, Phalcon, REST API

Databases: PostgreSQL, MySQL/MariaDB, MongoDB, Redis

Messaging: BullMQ, RabbitMQ, Apache Kafka

Security / Identity: Keycloak (SSO), OAuth2/OIDC, SpiceDB (ReBAC)

DevOps & Testing: Docker, GitLab CI, Jest, Playwright

EXPERIENCE

HONEYNET Joint Stock Company

Tech Lead & Full-stack Developer

2022 – Present

- Led the re-platforming of a file-storage and identity-management platform from PHP/Phalcon to **NestJS + Next.js**; the new version now serves **5,000+ students and lecturers**.
- Built a TR-069 network device management system that manages and controls **250,000+ devices** and processes information within seconds (PHP 8.3/Phalcon, MongoDB, RabbitMQ, GenieACS).
- Built IoT systems that process information within seconds: water-meter data ingestion in per-minute batches (MongoDB, RabbitMQ) and Python agents reading devices over Modbus at BTS sites.
- Built a Docker Compose development environment of ~20 services with **18 custom Dockerfiles** (PHP-FPM + Phalcon 3.4 → 5.9 side by side, PostgreSQL, Kafka, Keycloak, SpiceDB, MinIO).
- Adopted an AI-assisted development workflow (Claude Code, Codex) with lint/build/test gates and phase-based roadmaps; delivered an MVP (16 backend phases + e2e tests) in ~12 days.

Full-stack Developer

03/2016 – 2021

- Built a WiFi Controller starting 04/2018 with PHP/Phalcon and MariaDB (device management, RADIUS, monitoring, reporting, client API) that processes information within seconds; maintained it and upgraded it from PHP 7 to 8 through 2026.
- Developed an internal operations platform (CMS core) from 03/2020: COVID-19 safety declarations, QR codes, executive schedules, task management, Face ID integration and attendance reports; processes information within seconds; upgraded from PHP 7 to 8.

Freelance

2019 – 2022

- Maintained and fixed bugs in a primary-school management system used in **10+ provinces**.
- Built a plugin that manages call data from a PBX system for a company's internal management system.
- Built WordPress websites showcasing real-estate projects for several companies.

SELECTED PROJECTS

File Storage, Sharing & Identity Management Platform

07/2026 – Present

Tech Lead & Full-stack Developer · NestJS 11, Next.js 16, React 19, PostgreSQL, TypeORM, Redis, BullMQ, SpiceDB, Keycloak, MinIO, Kafka

- Developed all 5 repositories (2 backends, 3 frontends), **~106,000 lines of TypeScript** and 103 test files; the system serves **5,000+ students and lecturers**.
- Built folder trees with PostgreSQL **Itree**, trash, public links and owner/editor/viewer sharing on **SpiceDB**.
- Built a Keycloak administration backend via the Admin API (users, clients, roles, groups), including **bulk user import/update from Excel** with a review step and email tracking/resent.
- Split processes into http / worker / cron roles to scale the API and background jobs independently; synchronized MinIO bucket events through Kafka.
- Developed an embeddable **File Picker SDK** (iframe) for third-party apps, plus a demo app with SSO integration.

ACS Admin — TR-069 Network Device Management

04/2024 – 09/2026

Tech Lead & Full-stack Developer · PHP 8.3, Phalcon 5.6, MongoDB, MariaDB, GenieACS, RabbitMQ, Telegram Bot

- Delivered a platform that manages and controls **250,000+ devices** for a major cable TV & Internet provider, processing information within seconds.
- Built bulk configuration, diagnostics, network maps, device templates/tags and **condition-based alerts** via Telegram/email.
- Developed RabbitMQ workers for data synchronization and an integration API secured with secret keys + IP allowlisting (≈752 of 775 commits).

Digital Transformation Suite for a Ward-level Government Agency

07/2026 – 09/2026

Tech Lead & Full-stack Developer · NestJS 11, TypeORM, BullMQ, Socket.IO, Next.js 16, Auth.js v5, Keycloak SSO, Vitest, Playwright

- Built an IAM service for organizational master data with **two-way Keycloak synchronization** and a Directory API for other applications.
- Developed a paperless meeting system: schedules, documents, attendees and **real-time voting**; applied the Transactional Outbox pattern and controlled impersonation.
- Wrote **262 test files** across 3 backends (IAM 130, meetings 78, KPI 54) with Vitest and Playwright.

Face ID — Access Control & Attendance

07/2022 – 07/2026

Full-stack Developer · PHP/Phalcon, MongoDB, RabbitMQ, Redis, Firebase, Swagger

- Built a management system for face-recognition devices, door locks, snapshots and reports that processes information within seconds.
- Developed a **multi-tenant** version with event triggers, alert conditions and recognition schedules.

EDUCATION

Engineer's Degree — Faculty of Computer Networks and Communications

Graduated 2020

University of Information Technology — Vietnam National University, Ho Chi Minh City

LANGUAGES

English: reading technical documentation · Vietnamese: native